HYDROXYCARBAMIDE

Sickle cell disease

Teratogenic; FBC monitoring

CHEMICAL NAME Hydroxyurea (N-hydroxyurea, $CH_4N_2O_2$)
CLASS Antimetabolite (ribonucleotide reductase inhibitor)
TRADE NAMES Hydrea · Siklos · Droxia · Hydroxyurea (generic)

THERAPEUTIC USES / INDICATIONS

- Sickle cell disease (↓ painful crises, acute chest syndrome, transfusions, mortality) — adults and children.
- Myeloproliferative neoplasms: polycythaemia vera, essential thrombocythaemia.
- Chronic myeloid leukaemia (cytoreduction); head and neck cancer (with radiotherapy).

ADMINISTRATION

- **PO** once daily at the same time; swallow capsules whole with water.
- Caregivers wear gloves when handling; wash hands.
- Capsule contents not to be inhaled or touched.

DOSAGES

- **Sickle cell:** 15–20 mg/kg/day, increase by 5 mg/kg every 8–12 weeks to max tolerated dose (usually ≤35 mg/kg/day), guided by blood counts.
- **Polycythaemia/ET:** 15–20 mg/kg/day (usual 0.5–1.5 g/day).
- Reduce in renal impairment.

PHARMACOKINETICS

- **Peak:** 1–4 h. Good oral absorption; crosses BBB.
- Partly metabolised in liver.
- Excreted in urine. **t½:** 2–4 h.

PHARMACODYNAMICS

- Inhibits **ribonucleotide reductase** → ↓ deoxyribonucleotides → S-phase arrest.
- In sickle cell: ↑ fetal haemoglobin (HbF) → ↓ HbS polymerisation and sickling; ↓ neutrophils and adhesion.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, GI upset.
- Skin hyperpigmentation, nail changes, dry skin.
- Macrocytosis (expected).

ADVERSE EFFECTS (SERIOUS)

- Myelosuppression (neutropenia, thrombocytopenia); dose-limiting.
- Leg ulcers; skin cancer (long-term).
- Teratogenicity; reduced sperm count.
- Secondary leukaemia (uncertain, mainly in MPN).

CONTRAINDICATIONS & CAUTIONS

- Severe myelosuppression.
- Pregnancy, breastfeeding.
- Severe renal or hepatic impairment (caution).

KEY INTERACTIONS

- **Antiretrovirals (didanosine, stavudine):** pancreatitis, neuropathy.
- **Live vaccines:** contraindicated.
- **Other myelosuppressives:** additive.

NURSING MANAGEMENT

- FBC every 2–4 weeks during titration, then every 2–3 months; HbF levels.
- Monitor skin, leg ulcers.
- Contraception counselling.
- Sickle cell: hydration and crisis education.

PATIENT EDUCATION

- Take daily at the same time; handle capsules with gloves.
- Report fever, sore throat, bruising, leg ulcers.
- Use contraception; men may consider sperm banking.
- Benefit in sickle cell builds over months; continue.

KEY POINTS

- Ribonucleotide reductase inhibitor.
- Myelosuppression — regular FBC.
- Sickle cell: ↑ HbF; 15–35 mg/kg/day.
- Teratogenic; leg ulcers.